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Multiscale analysis of hypoxemia in methemoglobin anemia

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ABSTRACT

Methemoglobinemia is a disease that results from abnormally high levels of methemoglobin (*MetHb*) in the red blood cell (RBC), which is caused by simultaneous uptake of oxygen (O_2) and nitric oxide (NO) in the human lungs. *MetHb* is produced in the RBC by irreversible NO-induced oxidation of the oxygen carrying ferrous ion (Fe^{2+}) present in the heme group of the hemoglobin (Hb) molecule to its non-oxygen binding ferric state (Fe^{3+}). This paper studies the role of NO in the pathophysiology of methemoglobinemia and presents a multiscale quantitative analysis of the relation between the levels of NO inhaled by the patient and the hypoxemia resulting from the disease. Reactions of NO occurring in the RBC with both *Hb* and oxyhemoglobin are considered in conjunction with the usual reaction between oxygen and *Hb* to form oxyhemoglobin. Our dynamic simulations of NO and O_2 uptake in the RBC (micro scale), alveolar capillary (meso scale) and the entire lung (macro scale) under continuous, simultaneous exposure to both gases, reveal that NO uptake competes with the reactive uptake of O_2, thus suppressing the latter and causing hypoxemia. We also find that the mass transfer resistances increase from micro through meso to macro scales, thus decreasing O_2 saturation as one goes up the scales from the cellular to the organ (lung) level. We show that NO levels of 203 ppm or higher while breathing in room air may be considered to be fatal for methemoglobinemia patients since it causes severe hypoxemia by reducing the O_2 saturation below a critical value of 88%, at which Long Term Oxygen Therapy (LTOT) becomes necessary.

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1. Introduction

Methemoglobinemia is a disorder characterized by the presence of above-normal levels of methemoglobin (*MetHb*) in the blood. Methemoglobin is a compound formed from hemoglobin (*Hb*) by oxidation of iron atom from ferrous to ferric state, which is generated in the human respiratory system from the co-operative binding of hemoglobin with nitric oxide. *MetHb* lacks the electron needed to form a bond with oxygen (O_2), and hence, is incapable of oxygen transport. Thus, methemoglobinemia is characterized by moderate to severe hypoxemia. Case studies [1,2] reveal that when suffering from the disease, the arterial blood turns brown due to higher levels of *MetHb*, which produces a bluish tinge in skin color. Methemoglobinemia can be seldom linked to heredity and is most often of the acquired type, caused by external sources of nitric oxide [2]. Such sources include exposure to exogenous oxidizing drugs such as benzocaine, dapsone [3] and nitrate toxicity from contaminated water or industrial fumes. Greer and Shanon [4] in his clinical report on infant methemoglobinemia identified nitrate toxicity from water as one of the main reasons for infant methemoglobinemia. Such nitric oxide (NO) sources may accelerate the rate of formation of *MetHb* up to one-thousand fold, overwhelming the protective enzyme systems and acutely increasing susceptibility to the disease [5].

It may be mentioned that the chief agent of this disease, methemoglobin is not entirely foreign to the human body. It often arises from oxidant stresses continually undergone by RBCs, and hence is continually produced [6]. It is therefore instructive to take a closer look at the chemical reaction system of NO with the RBC hemoglobin in the presence of O_2, which is given by

$$Hb + O_2 \leftrightharpoons Hb(O_2)_n, \tag{1}$$

$$HbO_2 + NO \leftrightharpoons MetHb + NO_3^-, \tag{2}$$

$$Hb + NO \leftrightharpoons HbNO. \tag{3}$$

Reaction (1) shows that oxygen–*Hb* equilibrium is assumed to follow the Hill equation with *n*being the Hill constant (=2.34), and the equilibrium rate constant (K_1) being given by

$$K_1 = \left(\frac{H}{P_{50}}\right)^n, \tag{4}$$

where $H = 7.4 \times 10^5$ Torr and $P_{50} = 26$ Torr [7]. The reaction rate constant for NO induced oxidation of oxyhemoglobin to *MetHb* (given by Eq. (2)) was calculated by Eich et al. [8]. Scheele et al. [9] reported the reaction rate constant for the reaction between NO and deoxyhemoglobin (Eq. (3)) as 2×10^7 M^{-1} s^{-1}, and also

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measured the dissociation rate of nitrosylhemoglobin. The dissociation rate constant of *MetHb* (in Eq. (2)) is obtained to be 2.47×10^{-3} s^{-1} from Power et al. [10].

In normal RBCs, typical levels of methemoglobin produced are less than 1%. Its generation is controlled through two *MetHb* reducing pathways. One of these systems is the redox cycle consisting of cytochrome b5 (cytb5) and cytochrome b5-metHb reductase (b5R), which uses NADH for electron transfer to cytb5 ("cytb5-NADH system"). The other pathway uses flavin as an electron carrier for the reduction of *MetHb* coupled with NADPH oxidation, catalyzed by NADPH-dependent flavin reductase (FR) ("flavin NADPH system"). Both these major pathways have been discussed in great detail by Kinoshita et al. [11]. This enzymatic reduction of methemoglobin maybe written in a simplified way as

$$MetHb + MetHb - reductase(red) \leftrightharpoons Hb + MetHb(ox.). \tag{5}$$

It may be noted that Eq. (5) illustrates the conversion of *MetHb* back to normal hemoglobin and can thus be used in understanding the therapeutic strategies for the disease.

Thus, we see that this disease involves physical processes of gaseous exchange at the lung alveoli, diffusion and convection of dissolved gases in the alveolar capillaries and simultaneous chemical reaction and diffusion inside the red blood cells. Clearly, the phenomenon spans several varying length scales and may well be termed as a "*multiscale disease*". The scales involved are the micro-scale constituting the scale of the red blood cells, the meso-scale constituting capillary scale and the macro scale representing the scale of the organ, in this case, the whole lung. Transport at each of these levels, is coupled nonlinearly through the processes of diffusion, convection and reaction, the nonlinearity arising from the reaction kinetics of the participating gaseous components (in this case O_2 and NO) with hemoglobin. The smaller length scales are embedded hierarchically in the next larger length scale. Because of the coupling and hierarchical embedding of scales, the processes at smaller length scales significantly affect and alter those at larger scales. Clearly, any attempt to construct mathematical models of methemoglobinemia must be multiscale in nature and must take into account this intricate scale-coupling.

The literature reveals two major methodologies for solving multiscale transport–reaction problems in the human lung: the bottom-up and the top-down approaches. The *bottom-up approach* [12–14] relies on computationally intensive solution of detailed three dimensional convection–diffusion–reaction (CDR) equations without allowing any scope of approximation, while the *top-down approach* [15,16] makes a priori simplifying assumptions that often ignore scale coupling but offer analytical or semi-analytical solutions. Moreover, even in cases where detailed solutions are possible through the bottom up approach, the numerical results do not produce directly the results in terms of measurable macro-scale quantities (such as arterial/venous partial pressure of a gas), unless some averaging or coarse-graining is performed. Here we present the "averaging approach" which combines the mathematical rigor of the bottom-up approach with the computational ease of the top-down approach. Developed by Chakraborty and Balakotaiah [17], it consists of averaging/homogenizing the complete CDR equations over shorter length (time) scales to obtain low-dimensional models. Averaging the CDR equations and closing the averaged equations are performed by using the Liapunov–Schmidt (LS) technique of classical bifurcation theory. This averaging procedure is rigorous and can be done exactly to any order of required accuracy. Such low-dimensional models, unlike the conventional averaged models are characterized by two or more modes (depending on the number of length scales involved) instead of one. These modes are essentially representatives of the length (time) scales of the system and the interaction between them captures local or small-scale transport in the system. It must be pointed out that in spite of homogenization, the averaged equations preserve sufficient information about the short length (time) scales by retaining all the parameters and therefore, all the qualitative features of the full CDR equation. The LS technique also allows us to specify the region of validity of such low dimensional models [17]. The derivation and analysis of spatially averaged models for diffusion–reaction governed transport of oxygen in the RBC have been studied by Chakraborty et al. [18], while a multiscale low-dimensional approach to quantifying hypoxemia in Hepato-Pulmonary Syndrome (HPS) was presented by Chakraborty et al. [19]. Such an *averaging approach*combines the rigor of the bottom-up approach with the simplicity of the top-down approach.

In this paper, we attempt to quantify NO induced hypoxemia in methemoglobin anemia, with a particular focus on the more common acquired type of the disease. The reaction scheme outlined in Eqs. (1)–(4) provides us with the kinetics, which together with fundamental transport equations, constitute the entire model. As stated before, methemoglobinemia is essentially a *multiscale disease* with coupled transport and reaction spanning disparate length scales. In this work, we develop a rigorous multiscale model that consists of L–S averaged low dimensional forms of the fundamental CDR equations. The low-dimensional model for each scale is embedded hierarchically in the next larger scale and the model equations at all the three (micro, meso, macro) scales are solved not sequentially but simultaneously so as to allow coupling between the scales. We use our model to study the interdependent O_2 and NO dynamics in patients suffering from methemoglobinemia. Our model simulations allow us to quantify the increased methemoglobin levels inside the RBC in methemoglobinemia, as well as estimate the overall pulmonary saturations of oxygen and nitric oxide, that result in moderate to severe hypoxemia in the patient. Further, we also estimate the levels of inhaled NO that could be fatal for the patient and outline therapeutic strategies for the same.

2. Mathematical modeling

The multiscale models for quantification of hypoxemia in nitric oxide induced methemoglobinemia consist of developing CDR equations at the micro (RBC) scale and coupling it with similar transport models at the meso (capillary) and macro (lung) scales. In this section, the detailed governing equations at each scale are presented and the scale-coupling is demonstrated.

2.1. Micro scale: RBC

Transport of oxygen from the alveolus (where partial pressure of oxygen is denoted by P_{A,O_2}) to the red blood cell occurs across the membranes (alveolar epithelium and capillary endothelium) and the plasma. Human erythrocytes are biconcave discs with average dimensions of 8 mm in diameter and 1.6 mm in thickness.

2.1.1. Detailed model

The convection–diffusion–reaction equations in Lagrangian coordinates for a single RBC of any arbitrary shape with volume Ω and external surface area $\partial\Omega$ are given as

$$\alpha_j \frac{DP_{j,rbc}}{Dt} = D_j \alpha_j \nabla^2 P_{j,rbc} - R(S_i, P_{j,rbc}), \tag{6}$$

$$[Hb]_T \frac{DS_i}{Dt} = D_{Hb}[Hb]_T \nabla^2 S_j - R(S_i, P_{j,rbc}), \tag{7}$$

where $\frac{D}{Dt}$ is the substantial derivative and ∇^2 is the three dimensional Laplacian operator. $P_{j,rbc}$ is the partial pressure of the jth gaseous component dissolved in the RBC, S_i is the (fractional)

saturation of the ith species as obtained from the reaction scheme in Eqs. (1)–(3), D_j and D_{Hb} are the diffusion coefficients of the gaseous component j and Hemoglobin inside the erythrocyte respectively, $[Hb]_T$ is the total intra-cellular hemoglobin concentration and α_j is the solubility of the corresponding gaseous component in the RBC. The rates of reaction of the component gases with the hemoglobin in the RBC are functions of both partial pressure and saturation and are shown by $R(S_i, P_{j,rbc})$ in Eqs. (6) and (7). Considering a symmetric geometry, the appropriate boundary conditions for these equations are given by:

$$P_{j,rbc} \text{ and } S_i \text{ are finite at the center of the cell,} \tag{8}$$

$$(D_j\alpha_j\nabla P_{j,rbc})\cdot n = \eta(\langle P_{j,pl}\rangle - P_{j,rbc}) \text{ on } \partial\Omega \text{ (on the surface of the RBC),} \tag{9}$$

$$(\nabla S_i)\cdot n = 0 \text{ on } \partial\Omega \text{ (on the surface of the RBC).} \tag{10}$$

Eqs. (6) and (7) are also subject to initial conditions given by

$$P_{j,rbc}(t=0) = P_{V,O_2}, \tag{11}$$

$$S_i(t=0) = 0, \tag{12}$$

where n is the outward unit normal to the surface, P_{V,O_2} is the mixed venous partial pressure of oxygen, $\langle P_{j,pl}\rangle$ is the spatially averaged partial pressure of the jth gas component in the plasma. It has been shown [20] that a thin unstirred plasma layer is formed around the surface of the red cell and retards oxygen transfer from the plasma. η in Eq. (9) represents the mass transfer coefficient in the unstirred layer, which quantifies the mass transfer resistance between the RBC and the plasma. Solution of the coupled CDR equation of the RBC with the unstirred (boundary) layer surrounding it [18], gives

$$\eta = \left(\frac{D_{j,bl}\alpha_{j,bl}}{\delta}\right), \tag{13}$$

where δ is the thickness of the boundary layer, $\alpha_{j,bl}$ and $D_{j,bl}$ are the solubility and the diffusion coefficients of the corresponding component in the stagnant layer, respectively.

We proceed to write the system of equations in their detailed form where $j = 1$ and $j = 2$ refer to O_2 and NO, respectively, and $i = 1, 2$ and 3 indicate the species HbO_2, $MetHb$ and $HbNO$, respectively. We note here that in the rest of the paper we shall use the subscript i (j) interchangeably with the actual name of the species (gaseous component). In their complete form, the CDR equations for the micro scale, thus, are

$$\alpha_{O_2}\frac{\partial P_{O_2,rbc}}{\partial t} = D_{O_2}\alpha_{O_2}\nabla^2 P_{O_2,rbc} - R_1, \tag{14}$$

$$[Hb]_T\frac{\partial S_{HbO_2}}{\partial t} = D_{Hb}[Hb]_T\nabla^2 S_{HbO_2} + R_1 - R_2, \tag{15}$$

$$\alpha_{NO}\frac{\partial P_{NO,rbc}}{\partial t} = D_{NO}\alpha_{NO}\nabla^2 P_{NO,rbc} - (R_2 + R_3), \tag{16}$$

$$[Hb]_T\frac{\partial S_{MetHb}}{\partial t} = D_{Hb}[Hb]_T\nabla^2 S_{MetHb} + R_2, \tag{17}$$

$$[Hb]_T\frac{\partial S_{HbNO}}{\partial t} = D_{Hb}[Hb]_T\nabla^2 S_{HbNO} + R_3. \tag{18}$$

The spatially averaged partial pressure of the gas in the plasma that appears in the boundary conditions for this system represents the coupling that exists with the next higher scale, viz. the meso scale. Thus, to establish the complete low-dimensional scale coupled model, we spatially average Eqs. (14)–(18).

2.1.2. Spatially averaged model

We perform spatial averaging of the above equations over the volume of a single RBC using the Liapunov–Schmidt based spatial-averaging method [18,19], the detailed derivation for which has been presented in Appendix A. Post spatial-averaging, the micro-scale model for the simultaneous reactive uptake of O_2 and NO in RBC in Eulerian co-ordinates is obtained as

$$(\alpha_1 + [Hb]_T\beta_{11})\frac{D\overline{P_{1,rbc}}}{Dt} = \Theta_{11}(\overline{P_{1,pl}} - \overline{P_{1,rbc}}) + \Theta_{12}(\overline{P_{2,pl}} - \overline{P_{2,rbc}}), \tag{19}$$

$$[\alpha_2 + [Hb]_T(\beta_{22} + \beta_{32})]\frac{D\overline{P_{2,rbc}}}{Dt} = \Theta_{21}(\overline{P_{1,pl}} - \overline{P_{1,rbc}}) + \Theta_{22}(\overline{P_{2,pl}} - \overline{P_{2,rbc}}), \tag{20}$$

where, $\overline{P_{j,rbc}}$ is the spatially averaged partial pressure within the RBC, and β is the slope of the equilibrium dissociation curve given by $\beta_{ij} = \frac{\partial S_i}{\partial P_j}$. The quantity Θ_{ij} is called the 'diffusing capacity' [18,19], which, in a multi-component reaction–diffusion system, represents the reaction-altered mass transport coefficient of gas 'i' under the driving force of gas 'j'. For a single gaseous component, the diffusing capacity includes the effects of diffusional gradients within the RBC as well as that in the stagnant plasma layer, and the effect of facilitated transport of gas due to reactive coupling with hemoglobin. The expressions for the diffusing capacities are obtained as

$$\Theta_{11} = \frac{\alpha_1 + \beta_{11}[Hb]_T}{b\left(\frac{\alpha_1 D_1}{\eta_1} + \frac{b}{Sh_i}\right)}G_{11}, \tag{21}$$

$$\Theta_{12} = \frac{\alpha_1 + \beta_{11}[Hb]_T}{b\left(\frac{\alpha_2 D_2}{\eta_1} + \frac{b}{Sh_i}\right)}G_{12}, \tag{22}$$

$$\Theta_{21} = \frac{\alpha_1 + (\beta_{22} + \beta_{32})[Hb]_T}{b\left(\frac{\alpha_1 D_1}{\eta_2} + \frac{b}{Sh_i}\right)}G_{21}, \tag{23}$$

$$\Theta_{22} = \frac{\alpha_2 + (\beta_{22} + \beta_{32})[Hb]_T}{b\left(\frac{\alpha_2 D_2}{\eta_2} + \frac{b}{Sh_i}\right)}G_{22}, \tag{24}$$

where, η_i is a mass transfer coefficient of gas 'i' in the stagnant plasma layer surrounding the RBC [18], Sh_i is the internal Sherwood number of the RBC, and G_{ij} is a function of β_{ij}, the detailed expressions for which can be found in Appendix A.

An analytical expression for Sh_i for a discoid-shaped RBC as a function of its ratio of height (H) to diameter (D) has been given in Eq. (124) in Appendix C of [18]. Cellular deformations often alter H/D and the corresponding change in the Sherwood number may be calculated using the reference stated above. In general, erythrocytes have dimensions of 8 mm in diameter and 1.6 mm in thickness i.e. have a H/D of 0.2, and therefore, have $Sh_i = 2$ [18], which is the value used in our analysis.

2.2. Meso scale: capillary

A pulmonary capillary consists of a continuous plasma phase in which the red cells are suspended. Our meso-scale model assumes the plasma phase can be considered a continuum in which suspended RBCs behave like sources/sinks of O_2 and NO.

2.2.1. Detailed model

The meso scale model equations are given by

For oxygen:

$$(1-h)\left[\frac{\partial C_{O_2,pl}}{\partial t'}+v_{pl}(r,x)\frac{\partial C_{O_2,pl}}{\partial x}+R_{pl}(C_{O_2,pl})\right]+h\Theta_{11}(\overline{P_{O_2,pl}}-\overline{P_{O_2,rbc}})+h\Theta_{12}(\overline{P_{NO,pl}}-\overline{P_{NO,rbc}})=(1-h)D_{O_2,pl}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C_{O_2,pl}}{\partial r}\right)+\frac{\partial^2 C_{O_2,pl}}{\partial x^2}\right], \quad (25)$$

For nitric oxide:

$$(1-h)\left[\frac{\partial C_{NO,pl}}{\partial t'}+v_{pl}(r,x)\frac{\partial C_{NO,pl}}{\partial x}+R_{pl}(C_{NO,pl})\right]+h\Theta_{21}(\overline{P_{O_2,pl}}-\overline{P_{O_2,rbc}})+h\Theta_{22}(\overline{P_{NO,pl}}-\overline{P_{NO,rbc}})=(1-h)D_{NO,pl}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C_{NO,pl}}{\partial r}\right)+\frac{\partial^2 C_{NO,pl}}{\partial x^2}\right], \quad (26)$$

where h is the hematocrit i.e. volume fraction of RBC in the blood, $C_{j,pl}$ is the concentration of gas j ($j = 1$ or 2) in the plasma phase expressed in L (STPD) of O_2 per L blood, $D_{j,pl}$ is the effective diffusivity of gas in the plasma (effective diffusivity is used instead of molecular diffusivity, since it is known that the red cell rotation and other motions in the shear field tend to promote radial transport), Θ_{ij}s are given by Eqs. (21)–(24), $\overline{P_{j,rbc}}$ is obtained by solving the averaged equations in the micro scale (Eqs. (19) and (20)), and $C_{j,pl}$ and $P_{j,pl}$ are related linearly as

$$C_{j,pl}=\alpha_{j,pl}P_{j,pl}, \quad (27)$$

where $\alpha_{j,pl}$ is the solubility of gas j in the plasma. We may also note here, that $v_{pl}(r,x)$ is not equal to the velocity at which the red cells are carried along by the plasma, and there exists slip between the RBC and the plasma phase, which our model accounts for. Eqs. (25) and (26) are subject to the boundary conditions

$$\left(-\frac{2}{a}\right)(1-h)D_{j,pl}\frac{\partial C_{j,pl}}{\partial r}=\frac{D_{M,j}}{V_c}(P_{A,j}-P_{j,pl}|r=a) \quad \text{at } r=a, \quad (28)$$

$$\left.\frac{\partial C_{j,pl}}{\partial r}\right|_{r=0}=0, \quad (29)$$

where P_{A,O_2} is the alveolar partial pressure, $D_{M,j}$ is the membrane diffusing capacity of the jth component under consideration and has units of L (STPD) of O_2/(mmHg s), a is the capillary radius and V_C is the total capillary bed volume. Of the two boundary conditions, Eq. (29) ensures centerline symmetry, while Eq. (28) accounts for mass transfer across the membrane using the concept of membrane diffusing capacity and also couples the transport inside the capillary with that in the macro scale, i.e. the alveolus.

The velocity of the red cells (v_{rbc}) and the plasma (v_{pl}) are not constants but are given by [21,13]

$$v_{rbc}(r)=2(1-slip)\bar{v}_D(x)\left[1-B\left(\frac{r}{r_c}\right)^2\right], \quad r\in[0,r_c],$$

$$v_{pl}(r)=\begin{cases}2\bar{v}_D(x)\left[1-B\left(\frac{r}{r_c}\right)^2\right], & r\in[0,r_r]\\ 2\bar{v}_A(x)\left[1-\left(\frac{r}{r_c}\right)^2\right], & r\in[r_r,r_c]\end{cases},$$

while, the hematocrit, i.e., the volume fraction of the RBC at any radius is expressed as [21,13]

$$h=\begin{cases}2h_m\left[1-\left(\frac{r}{r_r}\right)^m\right], & r\in[0,r_r]\\ 0, & r\in[r_r,r_c]\end{cases}$$

where, r_r = radius of the RBC rich region, r_c = radius of the capillary, B = Blunting factor, that represents deviation from parabolic flow profile ($B\in[0,1]$), and *slip* = relative slip between plasma and RBC ($slip\in[0,1]$). However, for our analysis, we consider the hematocrit to be constant along the transverse direction (=h). We further assume a fully developed parabolic velocity profile for the plasma, given by

$$v_{pl}(r)=2\bar{v}_{pl}\left[1-\left(\frac{r}{a}\right)^2\right], \quad (30)$$

and a constant velocity profile for the RBC which retains the effect of slippage and is given by

$$v_{rbc}(r)=(1-slip)\bar{v}_{pl}, \quad (31)$$

where, $\bar{v}_{pl}$ = average velocity of plasma in the capillary. Typical values used later in this analysis are *slip* = 0.1 and h = 0.45. It may be further noted that since all reactions are essentially confined to the RBC and none occur in the plasma phase, $R_{pl}(C_{pl})=0$ in both Eqs. (25) and (26).

2.2.1.1. Non dimensionalization. In this section, we non-dimensionalize the meso-scale governing equations and their boundary conditions, in order to compare the magnitude of the various time scales involved in the transport process, and accordingly simplify the governing equations. The non-dimensional forms of the different variables are

$$t=\frac{t'}{\tau},\quad \zeta=\frac{r}{a},\quad z=\frac{x}{L},\quad u_{pl}=\frac{v_{pl}}{\overline{v_{pl}}}=2(1-\zeta^2),\quad u_{rbc}=\frac{v_{rbc}}{\overline{v_{pl}}}=(1-slip), \quad (32)$$

$$p_{j,pl}=\frac{P_{j,pl}}{P_{A,O_2}},\quad c_{j,pl}=\frac{C_{j,pl}}{\alpha_{j,pl}P_{A,O_2}},\quad p_{j,rbc}=\frac{P_{j,rbc}}{P_{A,O_2}}, \quad (33)$$

while the different dimensionless numbers are given by

$$Pe_{T,j}=\frac{a^2\overline{v_{pl}}}{LD_{j,pl}}=\frac{t_D}{\tau},\quad Pe_j=\frac{L\overline{v_{pl}}}{D_{j,pl}}=\frac{t_x}{\tau}, \quad (34)$$

$$\Theta_{rbc,ij}=\frac{L\Theta_{ij}}{v_{pl}\alpha_{j,pl}}=\frac{\tau}{t_{rbc-pl}},\quad \Theta_{j,m}=\frac{LD_{M,j}}{v_{pl}V_c\alpha_{j,pl}}=\frac{\tau}{t_{m-pl}}, \quad (35)$$

and the constituent time scales are given by

$$\tau=\frac{L}{v_{pl}},\quad t_D=\frac{a^2}{D_{pl}},\quad t_x=\frac{L^2}{D_{pl}},\quad t_{rbc-pl}=\frac{\alpha_{pl}}{\Theta},\quad t_{m-pl}=\frac{V_c\alpha_{pl}}{D_M}. \quad (36)$$

The above expressions show that the dimensionless numbers are ratios of different time-scales in the system, e.g. $Pe_{T,j}$ is the ratio of radial diffusion time to transit time of blood in the capillary (τ), Pe_j is the ratio of axial diffusion time to capillary transit time, $\Theta_{rbc,ij}$ is the ratio of capillary transit time to mass transfer time between RBC and plasma, while $\Theta_{j,m}$ is the ratio of blood transit time to mass transfer time across the alveolar membrane. Using these dimensionless variables, and using the fact that, $c_{j,pl}=p_{j,pl}$, the governing Eqs. (25) and (26) along with their boundary conditions can be written in the dimensionless form

For O_2 ($j=1$)

$$\frac{1}{\zeta}\frac{d}{d\zeta}\left(\zeta\frac{\partial p_{1,pl}}{\partial\zeta}\right)=Pe_{T,1}\left\{(1-h)\left(\frac{\partial p_{1,pl}}{\partial t}+u_{pl}\frac{\partial p_{1,pl}}{\partial z}-\frac{1}{Pe_1}\frac{\partial^2 p_{1,pl}}{\partial z^2}\right)+h\Theta_{rbc,11}(\overline{P_{1,pl}}-\overline{P_{1,rbc}})+h\Theta_{rbc,12}(\overline{p_{2,pl}}-\overline{p_{2,rbc}})\right\} \quad (37)$$

For NO ($j=2$)

$$\frac{1}{\zeta}\frac{d}{d\zeta}\left(\zeta\frac{\partial p_{2,pl}}{\partial\zeta}\right) = Pe_{T,2}\left\{(1-h)\left(\frac{\partial p_{2,pl}}{\partial t} + u_{pl}\frac{\partial p_{2,pl}}{\partial z} - \frac{1}{Pe_2}\frac{\partial^2 p_{2,pl}}{\partial z^2}\right) + h\Theta_{rbc,21}(\overline{P_{1,pl}} - \overline{P_{1,rbc}}) + h\Theta_{rbc,22}(\overline{p_{2,pl}} - \overline{p_{2,rbc}})\right\}, \tag{38}$$

with boundary conditions given by

$$\left.\frac{\partial p_{j,pl}}{\partial\zeta}\right|_{\zeta=1} = Pe_{T,j}\frac{\Theta_{j,m}}{2}(1 - p_{j,pl}|_{\zeta=1}), \tag{39}$$

$$\left.\frac{\partial p_{j,pl}}{\partial\zeta}\right|_{\zeta=0} = 0 \quad (\text{for } j = 1 \text{ and } 2). \tag{40}$$

An order of magnitude analysis of the convection, radial and axial diffusion times in a capillary shows that the radial diffusion time is the smallest, the convection time comes next, and the axial diffusion time is the largest. Since axial diffusion is very slow as compared to convection (i.e. Pe_1 is very large $\sim 10^3$), we neglect the axial diffusion term in the above model. We may also note here that the low value ($\sim 10^{-4}$) of the transverse Peclet number facilitates radial averaging of the above model. Similar to the process carried out in the micro-scale (RBC), the averaging at the meso-scale is also performed using Liapunov– Schmidt technique, which allows us to reduce the radial dimension of the model.

2.2.2. Spatially averaged model

The detailed process of the radial averaging of the meso-scale model has been presented in Appendix B, and here we shall directly present the averaged meso-scale model equations, represented by three modes, namely the averaged partial pressure of gas in the blood plasma $\overline{p_{i,pl}}$, the cup-mixing partial pressure of the gas, $p_{i,m}$ (which is the velocity-weighted average of the partial pressure), and the pressure inside the RBC, $\overline{p_{i,rbc}}$ – the micro-scale variable that affects the meso-scale due to scale-coupling. Post-averaging, we obtain a set of Differential Algebraic Equations (DAEs), consisting of two global evolution equations:

$$(1-h)\frac{dp_{1,m}}{dz} = \Theta_{1,m}(1 - p_{1,s}) - h\Theta_{rbc,11}(\overline{P_{1,pl}} - \overline{P_{1,rbc}}) - h\Theta_{rbc,12}(\overline{p_{2,pl}} - \overline{p_{2,rbc}}), \tag{41}$$

$$(1-h)\frac{dp_{2,m}}{dz} = \Theta_{2,m}(k - p_{2,s}) - h\Theta_{rbc,21}(\overline{P_{1,pl}} - \overline{P_{1,rbc}}) - h\Theta_{rbc,22}(\overline{p_{2,pl}} - \overline{p_{2,rbc}}), \tag{42}$$

and four local equations:

$$p_{1,m} = \overline{p_{1,pl}} - Pe_{T,1}\left[\frac{\Theta_{1,m}}{24}(1 - p_{1,s}) + \frac{(1-h)}{48}\frac{dp_{1,m}}{dz}\right], \tag{43}$$

$$p_{2,m} = \overline{p_{2,pl}} - Pe_{T,2}\left[\frac{\Theta_{2,m}}{24}(k - p_{2,s}) + \frac{(1-h)}{48}\frac{dp_{2,m}}{dz}\right], \tag{44}$$

$$p_{1,s} = \overline{p_{1,pl}} + \frac{Pe_{T,1}}{24}\left[3\Theta_{1,m}(1 - p_{1,s}) + (1-h)\frac{dp_{1,m}}{dz}\right], \tag{45}$$

$$p_{2,s} = \overline{p_{2,pl}} + \frac{Pe_{T,2}}{24}\left[3\Theta_{2,m}(k - p_{2,s}) + (1-h)\frac{dp_{2,m}}{dz}\right], \tag{46}$$

where $p_{j,s}$ is the steady-state partial pressure at the capillary-membrane interface, and the parameter k is the ratio between the alveolar partial pressures of O_2 and NO, whose value is $\sim 10^{-6}$ in this analysis. It may be mentioned that the effects of scale-coupling between the micro and the meso levels are manifested in the equations through the magnitude of k. The above system of DAEs must be solved in conjunction with the non-dimensional Eulerian forms of the micro scale Eqs. (19) and (20). The Eulerian forms of these equations are given as

$$\frac{(\alpha_1 + [Hb]_T\beta_{11})}{\tau}u_{rbc}\frac{d\overline{p_{1,rbc}}}{dz} = \Theta_{11}(\overline{P_{1,pl}} - \overline{P_{1,rbc}}) + \Theta_{12}(\overline{p_{2,pl}} - \overline{p_{2,rbc}}), \tag{47}$$

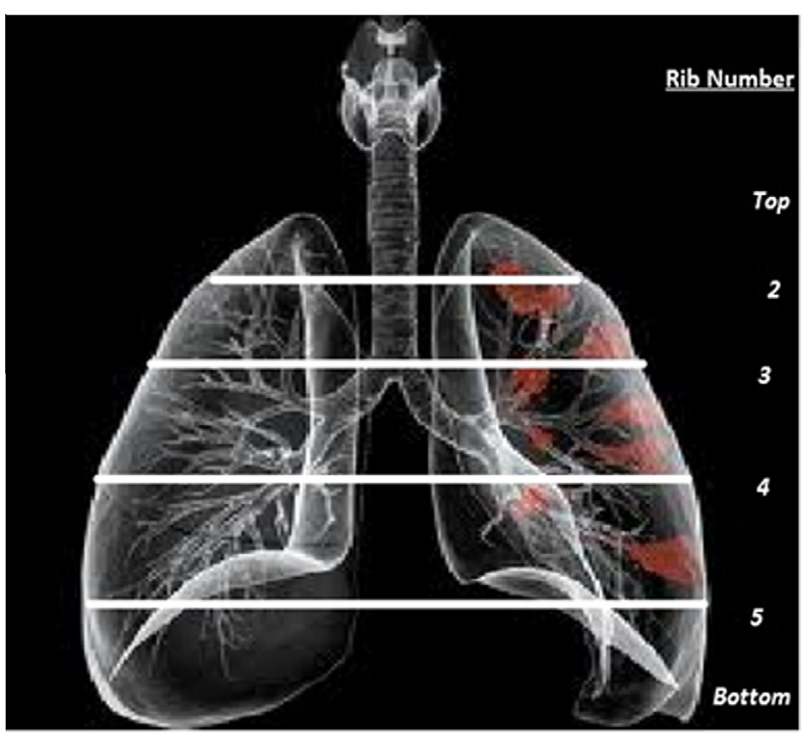

Fig. 1. Schematic of the lung showing positions of the compartments.

$$\frac{[\alpha_2 + [Hb]_T(\beta_{22} + \beta_{32})]}{\tau} u_{rbc} \frac{d\overline{p_{2,rbc}}}{dz} = \Theta_{21}(\overline{P_{1,pl}} - \overline{P_{1,rbc}}) + \Theta_{22}(\overline{p_{2,pl}} - \overline{p_{2,rbc}}). \quad (48)$$

2.3. Macro scale: lung

In this section, we develop relations governing gas ventilation ($\dot{V}$) and blood perfusion ($\dot{Q}$) in the lung, the ratio of which, known as the ventilation to perfusion ration ($\dot{V}/\dot{Q}$), controls the alveolar partial pressure (P_{A,O_2}), that in turn affects the meso and micro scales through scale coupling. For an "ideal" lung, this ratio is known to be 1 [22], but in a "normal" lung, it varies spatially across the different lung compartments, from 0.3 to 3.3. In addition, we also consider intrapulmonary right to left shunts ($\dot{V}/\dot{Q} = 0$), constituting about 3–5% of the blood flow. The spatial variation of ventilation and perfusion rates at the macro scale (i.e., lung) is accounted for by using a simple compartment model, which is illustrated in Fig. 1. As may be noted from Fig. 1, compartment 1 of our six compartment model corresponds to the space in the lung between ribs "Top" and rib number 2, compartment 2 corresponds to that between ribs 2 and 3, and so on, for the first five compartments. The fraction of blood flow through the sixth compartment corresponds to the shunt fraction f_{shunt}. For a "normal lung", the distribution of the ventilation to perfusion ratio with rib position has been obtained by fitting polynomials to data reported by West [22] (Fig. 6(A) in [18]), which is given as

$$\frac{\dot{V}}{\dot{Q}} = 0.3 + 3.24y - 7.19y^2 + 16.72y^4 - 21.54y^6 + 11.77y^8, \quad (49)$$

where y is the normalized distance along the ribs ($y \in [0, 1]$), given by

$$y = \frac{N_{rib,total} - N_{rib}}{N_{rib,total} - 1}, \quad (50)$$

where N_{rib} has been described in Fig. 3 and $N_{rib,total} = 6$. For the sixth compartment, i.e., the shunt, $\dot{V}/\dot{Q} = 0$. For the shunt, blood flow i.e. perfusion rate is given by $\dot{Q}_6 = \dot{Q}_{total} f_{shunt}$, while the remaining volume of flow is distributed across the other five compartments following a log-normal distribution, given by

$$\frac{\dot{Q}_{N_c}}{\dot{Q}_{total}(1 - f_{shunt})} = \Phi(N_{rib} = N_c) - \Phi(N_{rib} = N_c + 1), \quad (51)$$

where $N_c = 1, 2, 3, 4, 5$ is the compartment number and Φ is the cumulative distribution function of log-normal distribution, given by

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{x^2}{2}\right) dx, \quad (52)$$

$$\text{and } x = \frac{1}{\sigma}\left[\ln\left(\frac{\dot{V}}{\dot{Q}}\right)\bigg|_{N_{rib}} - \mu\right]. \quad (53)$$

As mentioned earlier, the alveolar partial pressure of O_2, P_{A,O_2} is different for each compartment and is determined by the local $\dot{V}/\dot{Q}$ ratio of that compartment. The relationship between P_{A,O_2} and the venous partial pressure of O_2 denoted by P_{v,O_2}, has been fitted as a sigmoidal function of $\dot{V}/\dot{Q}$ to the data presented by West [22] and the relationship is given by

$$P_{A,O_2} = P_{v,O_2} + c_1 \frac{(\dot{V}/\dot{Q})^{c_2}}{1 + (\dot{V}/\dot{Q})^{c_2}}. \quad (54)$$

For a "normal lung" breathing in room air, $P_{v,O_2} = 40$ mmHg and we obtain from the data, $c_1 = 122.5$ mmHg and $c_2 = 1.28$. It may be mentioned here, that the $\dot{V}/\dot{Q}$ value used in Eq. (53) is an average value in that particular compartment, obtained from integrating Eq. (48) as

$$\left\langle \frac{\dot{V}}{\dot{Q}} \right\rangle_{N_c} = \frac{\int \left(\frac{\dot{V}}{\dot{Q}}\right)(y) dy}{\int dy}, \quad (55)$$

where the integral is taken over the whole compartment N_c. Writing out the limits appropriately, the above quantity becomes

$$\left\langle \frac{\dot{V}}{\dot{Q}} \right\rangle_{N_c} = (N_{rib,total} - 1) \int_{\frac{N_{rib,total} - N_c - 1}{N_{rib,total} - 1}}^{\frac{N_{rib,total} - N_c}{N_{rib,total} - 1}} \left(\frac{\dot{V}}{\dot{Q}}\right)(y) dy. \quad (56)$$

For the sixth compartment, as mentioned earlier the ventilation–perfusion ratio is simply zero. Using the values of P_{A,O_2} and $\dot{V}/\dot{Q}$ for each compartment obtained by the above method, we calculate the partial pressure of the gas (where O_2 is of chief interest but we also calculate for NO) and the saturation of blood exiting each compartment (S_{a,N_c,O_2} and $S_{a,N_c,NO}$, respectively) using the steady-state meso-scale model (Eqs. (41)–(46)). Blood from the six compartments is allowed to mix in the pulmonary vein, and this mixing process is modeled as one in a perfectly stirred tank. The final gas saturation of mixed blood in the pulmonary vein, ($S_{a,j}$) determines the total amount of uptake of the gaseous component j (=1 for O_2 and 2 for NO) in the lung, which is given by

$$S_{a,j} = S_{v,j} f_{shunt} + \left(\frac{1}{\dot{Q}_{total}}\right) \sum_{N_c=1}^{5} \dot{Q}_{N_c} S_{a,N_c,j}, \quad (57)$$

where $S_{v,j}$ is the venous saturation of the gas. For O_2, this is calculated from the venous partial pressure, using the relation proposed by Severinghaus [23],

$$S_{v,O_2} = \frac{1}{1 + 23\,400(P_{v,O_2}^3 + 150 P_{v,O_2})^{-1}}, \quad (58)$$

while for NO,

$$S_{v,NO} = 0. \quad (59)$$

Thus, we have established a hierarchy of scales where the macro scale embeds the meso and the meso scale embeds the micro scale, and the entire system must be solved simultaneously (not sequentially) to capture the effect of the transport–reaction coupling across the scales.

2.4. Solution of coupled micro, meso and macro scales

Considering the macro scale compartmental model with the micro and meso scale ODEs, we have a system of coupled DAEs. However, numerical solution of this system in its present form results in an ill-conditioned mass matrix that is hard to handle, even with standard tools such as MATLAB. So, we exploit the linear interdependency of the modes in the meso-scale local equations (Eqs. (43)–(46) to convert the system of DAEs into a system of 4 coupled ODEs that describe the coupling at the two lower scales in terms of the partial pressure in the RBC (which is representative of the micro-scale) and the cup-mixing partial pressure in the plasma (which is representative of the meso-scale). The macro scale algebraic equations supply the compartmental alveolar partial pressure (Eq. (54)) as a boundary condition to this micro-meso system of ODEs. For each compartment, the ODE system computes the saturation and partial pressure profiles which are then combined together by way of Eq. (57) to obtain macro scale saturations. The entire schematic is presented in the form of an information flow diagram in Fig. 2 that reflects the numerical coupling between and hierarchical embedding of scales. It may be mentioned that the primary objective of solving the macro scale and obtaining the re-

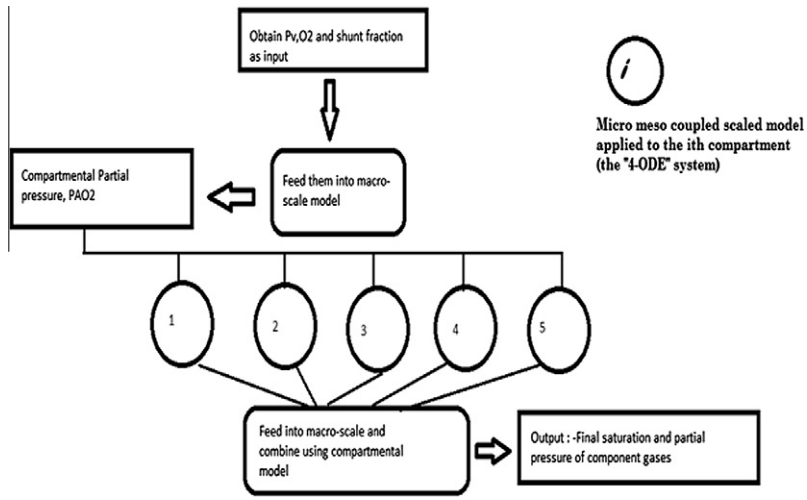

Fig. 2. Information flow diagram, demonstrating how variables communicate across different scales, and how the simulation algorithm functions.

sults in terms of the macro-scale variables is that these are most easily measurable and allows us to perform a host of parametric studies, which bring out interesting dependencies, as shall be seen later.

The derivation of the final ODE system fit for numerical solution, involves only elementary algebra (details in Appendix C) and here we directly present the final form as

$$X_1 \frac{dp_{1,m}}{dz} + Y_1 \frac{dp_{2,m}}{dz} = Z_1, \tag{60}$$

$$X_2 \frac{dp_{1,m}}{dz} + Y_2 \frac{dp_{2,m}}{dz} = Z_2, \tag{61}$$

$$\overline{X}_1 \frac{dp_{1,m}}{dz} + \overline{Y}_1 \frac{dp_{2,m}}{dz} + \frac{dp_{1,rbc}}{dz} = \overline{Z}_1, \tag{62}$$

$$\overline{X}_2 \frac{dp_{1,m}}{dz} + \overline{Y}_2 \frac{dp_{2,m}}{dz} + \frac{dp_{2,rbc}}{dz} = \overline{Z}_2, \tag{63}$$

where the coefficients are essentially the original variables as in Eqs. (41)–(48), grouped suitably so as to produce this system that lends itself to direct numerical solution using standard software.

The solution of the system of ODEs (Eqs. (60)–(63)) was performed using MATLAB. The mass matrix being extremely stiff and ill-conditioned due to the large difference in variable values across different scales, appropriate stiff solvers were employed.

3. Results and discussion

In this section, we present the O_2 and NO saturation (in hemoglobin) profiles that quantify gaseous uptake in methemoglobinemia, as well as study the effects of different parameters (including NO concentration) on the onset of the disease by simulating the multiscale model derived in the previous section. Table 1 lists the standard data set used for the simulation, along with the equivalent of the units used, in S.I. It may be mentioned here, that the diffusing capacity for O_2 transport through the capillary membrane ($D_{M,O2}$) is known from literature [18], while that for NO is estimated using the formula

$$\frac{D_{M,NO}}{D_{M,O_2}} = \frac{\alpha_{NO,pl} D_{NO,membrane}}{\alpha_{O2,pl} D_{O_2,membrane}}, \tag{64}$$

where $D_{j,membrane}$ is the membrane diffusivity of the gas j in the RBC. The membrane diffusivities for both O_2 and NO are obtained from available literature [24,25], and are listed in Table 1. We use deoxygenated blood with an O_2 partial pressure of 40 Torr and $S_{v,O_2} = 75\%$ as initial conditions for our simulation.

Table 1
Data used in simulation.

Variable	Value
K_1	2.648×10^{10} Torr^{-1}
K_2	2.1×10^{9} Torr^{-1}
K_3	3.4×10^{9} Torr^{-1}
n	2.34
$\alpha_{O_2,rbc}$	$1.56 \times 10^{-6} \frac{\text{mol}}{\text{L mmHg}}$
$\alpha_{NO,rbc}$	$2.58 \times 10^{-6} \frac{\text{mol}}{\text{L mmHg}}$
$\alpha_{O_2,pl}$	$1.3 \times 10^{-6} \frac{\text{mol}}{\text{L mmHg}}$
$\alpha_{NO,pl}$	$2.15 \times 10^{-6} \frac{\text{mol}}{\text{L mmHg}}$
D_{O_2}	9.5×10^{-6} cm^2/s
D_{NO}	3.5×10^{-5} cm^2/s
D_{Hb}	1.44×10^{-7} cm^2/s
$D_{O_2,pl}$	2.5×10^{-5} cm^2/s
$D_{NO,pl}$	4.79×10^{-5} cm^2/s
$D_{O_2,membrane}$	0.727×10^{-5} cm^2/s
$D_{NO,membrane}$	0.4×10^{-5} cm^2/s
V_c	60 mm
D_{M,O_2}	40 Torr^{-1} min^{-1}
$D_{M,NO}$	36.4 Torr^{-1} min^{-1}
P_{V,O_2}	40 mmHg
Hb_T	20.3×10^{-3} mol/l
h	0.45
δ	0.75×10^{-4} cm
a	4×10^{-4} cm
b	0.8×10^{-4} cm
Sh_i	2
L	0.1 cm
$slip$	0.1
τ	1 s
μ	0
σ	0.34
c_1	122.5 mmHg
c_2	1.28
$N_{rib,total}$	6

The equivalent units in S.I are 1 mmHg = 133.2 Pa, 1 mm = 10^{-3} m, 1 cm = 10^{-2} m, 1 l = 10^{-3} m^3, 1 Torr = 1 mmHg = 133.2 Pa, 1 min = 60 s.

Fig. 3 shows the variation of the micro scale diffusing capacities with the O_2 partial pressure in the RBC, where the NO concentra-

tion in the RBC is a parameter. As was briefly stated in Section 2.1.1 these diffusing capacities represent the reaction-enhanced effective diffusion coefficients of one gas in the presence of another. Fig. 3(a) shows the diffusing capacity of O_2 plotted against the partial pressure of O_2 in the RBC. The biphasic nature of the plot shows that O_2 uptake decreases for low or very high partial pressures and maximizes at around 25 Torr. Fig. 3(b) and (c) shows Θ_{12} and Θ_{21}, which are indicative of the effect of gas 1 on 2 and vice versa and reveal that they have nearly the same magnitude. This shows that the change O_2 and NO enforce upon the diffusion dynamics of each other is almost similar in extent. However, as seen in Fig. 3(d), Θ_{22} is several orders of magnitude higher than the rest of the capacities, which easily shows why NO is so potent. The reaction altered diffusion of NO, thus, is extremely high and any effect O_2 has in stemming it is considerably small. This is where we can begin to infer that even small quantities of NO, in finite time, must lead to the onset of methemoglobinemia if not removed or treated properly.

Fig. 4 illustrates the results of the meso scale simulation. The saturations of the species involved, O_2, NO, methemoglobin (*MetHb*) and nitrosyl-hemoglobin (*HbNO*) are plotted vs. the alveolar NO concentrations. Since this set of plots is obtained at the meso-scale, the compartmental variations of P_{A,O_2} do not affect the values significantly. Hence, P_{A,O_2} has been kept as a parameter and values assigned to it are nearly equal to the actual compartmental values that result from a macro-scale calculation. As is intuitively obvious, increasing P_{A,O_2} increases the critical NO concentration above which O_2 saturation falls below 90%, leading to severe hypoxemia. Also, the variation in NO saturation is significantly less than that for O_2, which bolsters our claim about the inability of the O_2 environment to stem NO dynamics. This effect is more pronounced at the macro scale as we shall see later. We may also observe from Fig. 4(b)–(d), that the principal contribution to the NO saturation is from the generated methemoglobin and only very little from *HbNO*. This can be interpreted as an indirect indication of the greater role of *MetHb* in causing metheoglobinemia (and not *nitrosylhemoglobin-anemia* for instance). Indeed, NO concentrations of just 60 ppm or higher in the lung can cause methemoglobin concentrations to spike (at P_{A,O_2} values of 80–90 Torr). Methmoglobin levels steadily rise from 1% (in disease-free condition) to as high as 10%, with increase in NO concentration, leading to the onset of methemoglobinemia (as corroborated by others [5]).

Fig. 5 presents the O_2 and NO saturation profiles in the macro-scale. The results obtained at this scale are the final outputs of the entire multi-scale simulation, spanning the micro, meso and macro scales, and are reported in terms of measurable physiological quantities at the macro-scale. Here, the parameter chosen is the venous partial pressure of O_2, which is varied from 20 Torr to 40 Torr. It may be noted that the macro-scale saturations of both

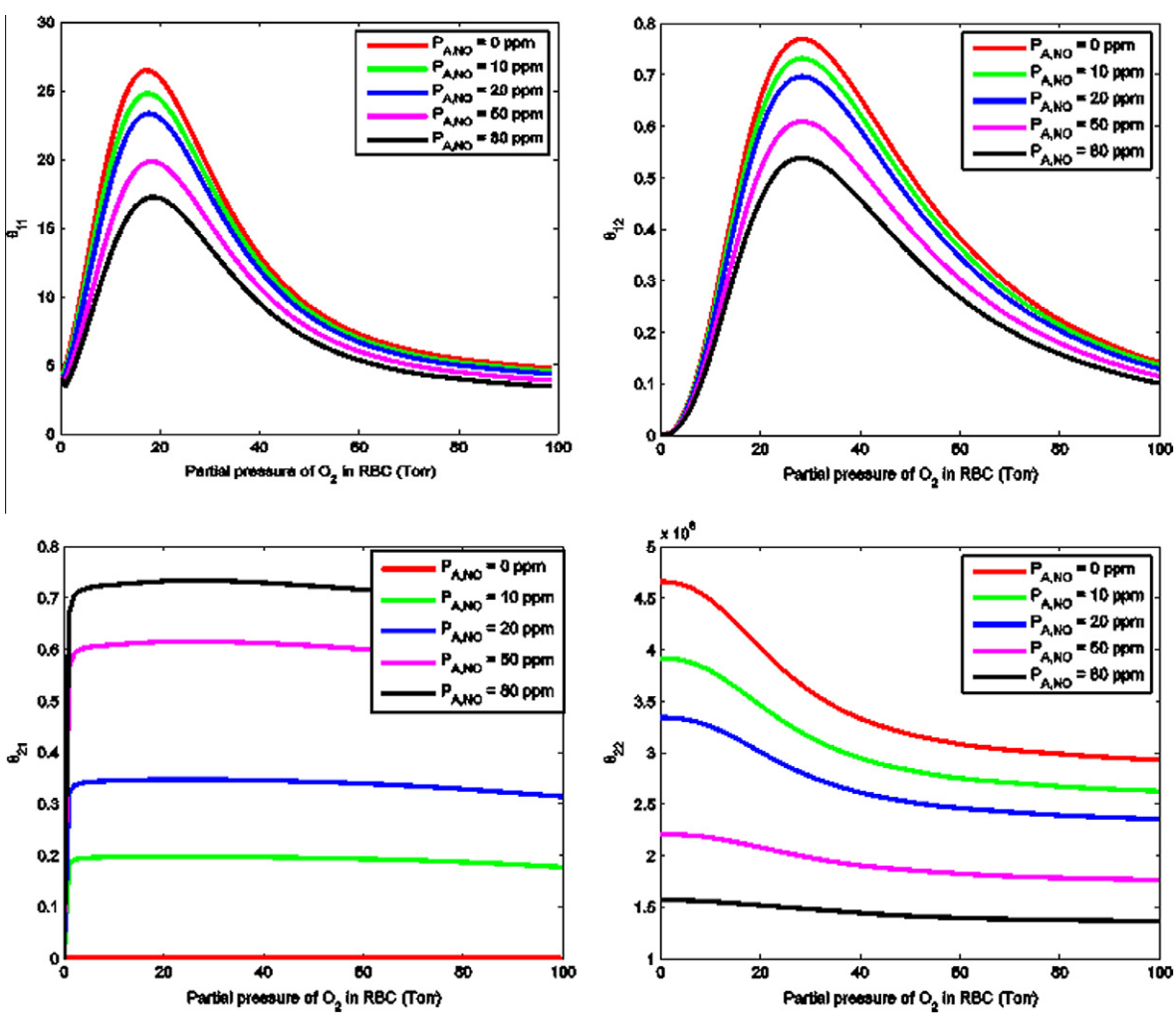

Fig. 3. Plots of micro scale diffusive capacities against partial pressure of oxygen in the RBC, with alveolar partial pressure of NO as a parameter.

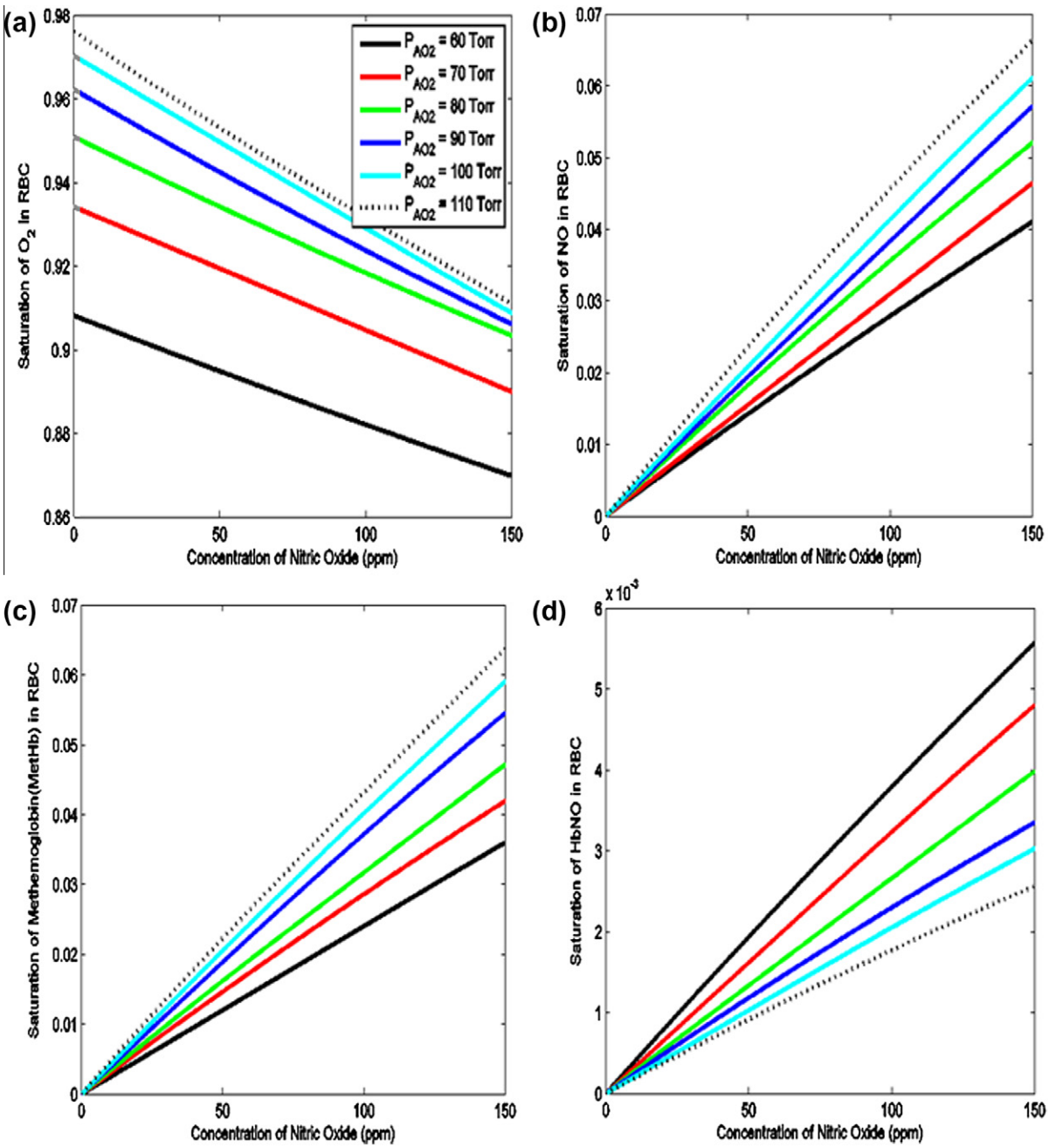

Fig. 4. Saturation profiles for (a) oxygen, (b) nitric oxide, (c) methemoglobin and (d) nitrosyl hemoglobin in the capillary (meso-scale) for different values of alveolar partial pressure of oxygen.

O_2 and NO have diminished significantly from those reported at the meso-scale. This happens because as we embed smaller scales hierarchically in larger ones their mass transfer resistances add up (not necessarily linearly). The increase in total resistance is observed to increase significantly when the macro scale is added to the coupled micro–meso scales. We also find that the variation in NO saturation in the macro scale is lesser than the corresponding variation in the meso scale while O_2 saturation gets considerably affected by alveolar NO concentrations in both these scales. Since macro scale results represent directly measurable quantities, we can claim that the physical conditions of O_2 do not have significant influence on the NO dynamics, while NO deposition in the alveolus considerably alters O_2 dynamics. Our present conclusion, in conjunction with the effect of the very high diffusing capacity Θ_{22} (Fig. 3(d)) shows why methemoglobinemia occurs; – NO keeps accumulating in the alveolar epithelium undergoing unhindered reactive uptake, while it keeps competing with O_2 uptake, causing subsequent hypoxemia.

In Fig. 6, the species saturations vs. NO levels are presented again, but this time with intrapulmonary left to right shunt fraction as a parameter. Two values are considered: a zero shunt and a standard 5% shunt. As is expected, saturations are higher for the zero shunt case since this allows more blood supply to oxygenated alveolar regions. Once again we find that NO saturations do not change much for different shunt fractions due to the very high reactivity (Θ_{22}) of NO. As mentioned before, the macro-scale has been modeled as a series of compartments with all the intrapulmonary shunt lumped together into the 6th (and last) compartment.

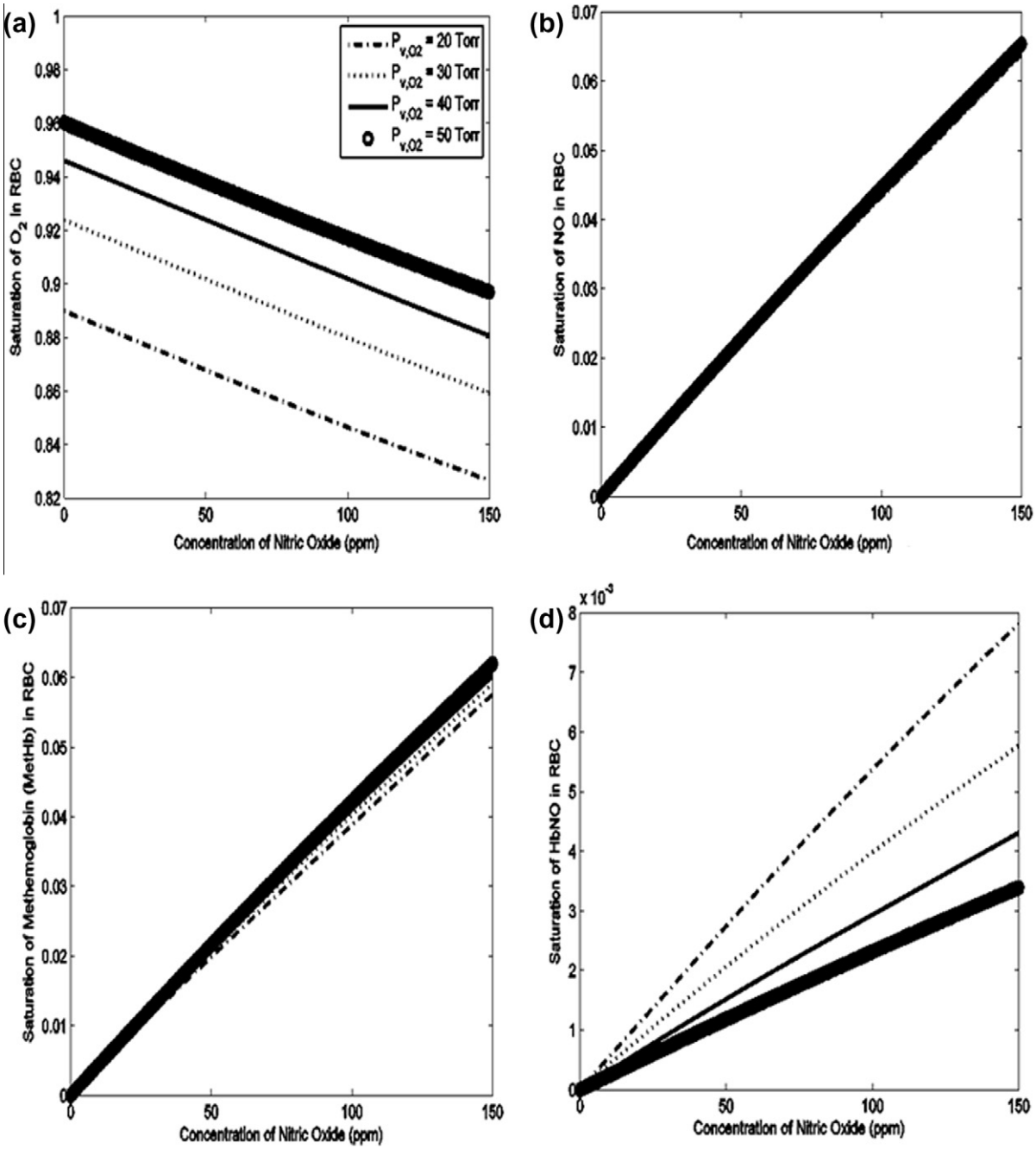

Fig. 5. Saturation profiles for (a) oxygen, (b) nitric oxide, (c) methemoglobin and (d) nitrosyl hemoglobin in the lung (macro-scale) for different values of venous partial pressure of oxygen.

When these compartments, with their different P_{A,O_2}s are solved in the macro-scale computations, high reactivity of NO ensures significant total NO uptake even before the last compartment, i.e. the shunt, is reached. Thus, whether we simulate with a 5% shunt or a zero shunt, when the saturations obtained from each compartment are combined, the final result has little to do with the shunt used, as is obvious from Fig. 6(a) and (b). Also, From Fig. 6(a) we find that even with zero shunt, the O_2 saturation tends to dip below the critical value of 90% at NO levels slightly higher than 150 ppm.

Finally, in Fig. 7 we present the variation of critical NO concentrations plotted against the venous partial pressure of O_2. As mentioned before, competition between the reactive uptakes of the two gases, suppresses O_2 use, and leads to hypoxemia. Eighty-eight percentage O_2 saturation is usually termed as the critical O_2 saturation at which Long Term Oxygen Therapy (LTOT) becomes necessary. (as per ATS guidelines [26]). Drop in saturation below 88% causes severe hypoxemia. Our simulations reveal that for the standard case of 40 Torr P_{v,O_2} and 5% shunt, the critical concentration of NO is around 203 ppm. Also, we see that the critical ppm increases with P_{v,O_2} and shunt fraction. A higher critical concentration of NO in a methemoglobinemia patient implies less susceptibility to extreme hypoxemia. For NO levels close to and beyond the critical, the patient needs to be treated with methylene blue, which works as a cofactor for the enzyme NADPH-methemoglobin reductase that helps in rapid conversion of *MetHb* back to *Hb*. To simulate the therapeutic process, our initial kinetic model would need to include Eq. (5) (in addition to Eqs. (1)–(3) and the CDR model (Eqs.

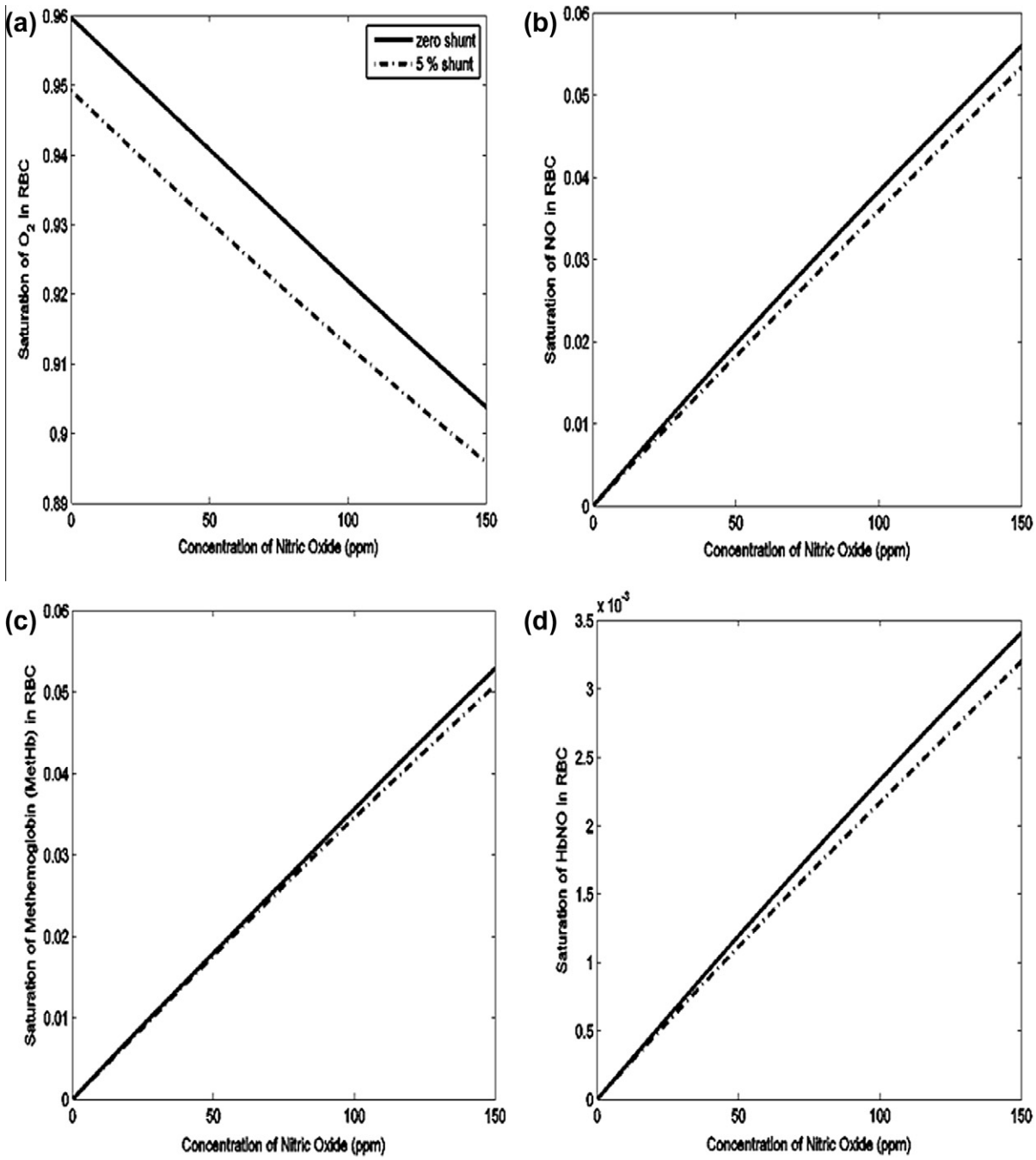

Fig. 6. Saturation profiles for (a) oxygen, (b) nitric oxide, (c) methemoglobin and (d) nitrosyl hemoglobin in the lung (macro-scale) for different values of shunt and maintaining venous partial pressure of oxygen at 40 Torr.

(14)–(18) in the micro scale and Eqs. (37)–(40) in the macro scale) and the subsequent spatially averaged low dimension analyses would have to be modified accordingly. The dynamics of the therapy of methemoglobinemia shall be pursued in our future publications.

4. Conclusions

In this analysis, we develop a complete multi-scale model of methemoglobin anemia and provide quantitative estimates of different important quantities that might improve our understanding of the disease and its possible therapeutic strategies. We start with the fundamental convection–diffusion–reaction equations and derive a coupled three (micro–meso–macro) scale model for the reactive transport of nitric oxide and oxygen in the red blood cell, the pulmonary capillary and the lung. The multi-scale model is subjected to a rigorous Liapunov–Schmidt reduction (of the classical Bifurcation Theory) which reduces the dimensionality of the model, thus reducing the computational effort for solution while retaining all the essential physics of the original CDR model. The low dimensional model of each scale is embedded hierarchically in the next larger scale and the models at the three scales are solved simultaneously. Numerical simulation of our coupled multiscale model provides us with quantities of physiological significance and direct insights into the mechanism of methemoglobinemia. By way of comparing the saturation profiles

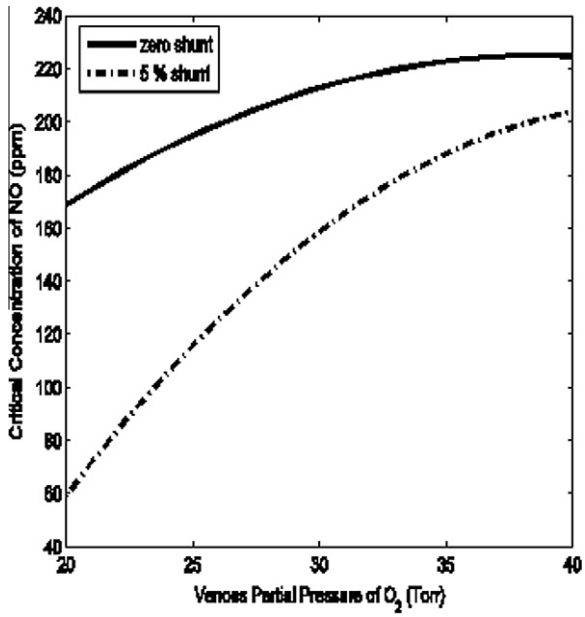

Fig. 7. Critical nitric oxide levels (in ppm) for different values of venous partial pressure of oxygen simulated at zero and 5% shunt fractions.

of O_2 and NO at different length scales, we show that while the presence of NO significantly influences reactive uptake of O_2, the O_2 partial pressure hardly affects NO deposition in the alveolar epithelium. This is central to the understanding of the why and the how of the disease.

The multi-scale nature of the disease is reflected in the reduction of overall gas saturations (both O_2 and NO) as we pass from one scale to the next higher one. Mathematically, it illustrates the intricate scale coupling, while physically it reflects the increased mass transfer resistance across the scales. One of the key findings of this paper is the critical or the safe limit of NO in the lung alveoli. Our simulations show that for NO concentration above 203 ppm, O_2 saturation falls below 88%, at which severe hypoxemia occurs necessitating the administration of Oxygen therapy. Thus, we find that an alveolar NO concentration of 203 ppm or higher in methemoglobinemia patients would be fatal due to severe hypoxemia and requires immediate therapy. As a natural extension, we go on to briefly discuss the commonly-used therapeutic approach for methemoglobinemia and outline the changes that need to be incorporated in our kinetic and transport models to quantify the dynamics of the therapeutic process using a similar multiscale analysis. We hope that this work shall help us evolve a multiscale methodology for quantifying other multiscale physiological disorders.

Appendix A. Diffusing capacities and spatial averaging in the micro-scale

In this section we present the detailed expressions for the diffusing capacities introduced in Section 2.1.2. To simplify the notation, we use the subscripts 1, 2 and 3 to denote saturations of species HbO_2, *MethHb* and *HbNO*, respectively, and P_1 and P_2 to denote $P_{1,rbc}$ and $P_{2,rbc}$, respectively. Treating the reaction scheme in Eqs. (1)–(3) as elementary, the equilibrium relations in terms of fractional saturations can be written as

$$K_1 = \frac{S_1}{S_{Hb}\alpha_1^n P_1^n}, \tag{A.1}$$

$$K_2 = \frac{S_2}{S_1\alpha_2 P_2}, \tag{A.2}$$

$$K_3 = \frac{S_2}{S_{Hb}\alpha_2 P_2}, \tag{A.3}$$

Also, at any instant the total hemoglobin in the system (free + - bound form) is constant. The total hemoglobin saturation in the RBC is 100%, i.e.,

$$S_1 + S_2 + S_3 + S_{Hb} = 1. \tag{A.4}$$

Solving these above equations, we have the fractional saturations as

$$S_1 = \frac{K_1\alpha_1^n P_1^n}{1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2}, \tag{A.5}$$

$$S_2 = \frac{K_1K_2\alpha_1^n\alpha_2 P_1^n P_2}{1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2}, \tag{A.6}$$

$$S_3 = \frac{K_3\alpha_2 P_2}{1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2}. \tag{A.7}$$

The dissociation constants, as mentioned in Section 2.1.2, are obtained by differentiation and are given as

$$\beta_{11} = \frac{\partial S_1}{\partial P_1} = \frac{K_1\alpha_1^n(1 + K_3\alpha_2 P_2)nP_1^{n-1}}{(1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2)^2}, \tag{A.8}$$

$$\beta_{12} = \frac{\partial S_1}{\partial P_2} = \frac{-K_1\alpha_1^n(K_3\alpha_2 + K_1K_2\alpha_1^n\alpha_2 P_1^n)P_1^n}{(1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2)^2}, \tag{A.9}$$

$$\beta_{21} = \frac{\partial S_2}{\partial P_1} = \frac{K_1K_2\alpha_1^n\alpha_2 P_2(1 + K_3\alpha_2 P_2)nP_1^{n-1}}{(1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2)^2}, \tag{A.10}$$

$$\beta_{22} = \frac{\partial S_2}{\partial P_2} = \frac{K_1K_2\alpha_1^n\alpha_2(1 + K_1K_2\alpha_1^n P_1^n)P_1^n}{(1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2)^2}, \tag{A.11}$$

$$\beta_{31} = \frac{\partial S_3}{\partial P_1} = \frac{-K_3\alpha_2 P_2(K_1\alpha_1^n + K_1K_2\alpha_1^n\alpha_2 P_2)nP_1^{n-1}}{(1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2)^2}, \tag{A.12}$$

$$\beta_{32} = \frac{\partial S_3}{\partial P_2} = \frac{K_3\alpha_2(1 + K_1\alpha_1^n P_1^n)}{(1 + K_1\alpha_1^n P_1^n + K_1K_2\alpha_1^n\alpha_2 P_1^n P_2 + K_3\alpha_2 P_2)^2}. \tag{A.13}$$

Time derivatives in Eqs. (14)–(18) are written in terms of β_{ij}s as

$$\frac{\partial S_i}{\partial t} = \frac{\partial S_i}{\partial P_{1,rbc}}\frac{\partial P_{1,rbc}}{\partial t} + \frac{\partial S_i}{\partial P_{2,rbc}}\frac{\partial P_{2,rbc}}{\partial t} = \beta_{i1}\frac{\partial P_{1,rbc}}{\partial t} + \beta_{i2}\frac{\partial P_{2,rbc}}{\partial t}, \tag{A.14}$$

and then the reaction rate terms are eliminated from Eqs. (14)–(18) to obtain

$$\frac{dP_1}{dt} = G_{11}\nabla^2 P_1 + G_{12}\nabla^2 P_2, \tag{A.15}$$

$$\frac{dP_2}{dt} = G_{21}\nabla^2 P_1 + G_{22}\nabla^2 P_2, \tag{A.16}$$

where the coefficients are given as

$$\left\{\begin{array}{l} G_{11} = [D_1\alpha_1\alpha_2 + [Hb]_T\{D_1\alpha_1(\beta_{22} + \beta_{32}) + D_{Hb}\alpha_2(\beta_{11} + \beta_{21})\} + \\ \qquad D_{Hb}[Hb]_T^2(\beta_{11}\beta_{22} + \beta_{11}\beta_{32} + \beta_{21}\beta_{32} - \beta_{21}\beta_{12} - \beta_{31}\beta_{12} - \beta_{31}\beta_{22})]/Deno; \\ G_{12} = [Hb]_T\alpha_2(\beta_{12} + \beta_{22})(D_{Hb} - D_2)/Deno; \\ G_{21} = [Hb]_T\alpha_1(\beta_{21} + \beta_{31})(D_{Hb} - D_1)/Deno; \\ G_{22} = [D_2\alpha_1\alpha_2 + [Hb]_T\{D_{Hb}\alpha_1(\beta_{22} + \beta_{32}) + D_2\alpha_2(\beta_{11} + \beta_{21})\} + \\ \qquad D_{Hb}[Hb]_T^2(\beta_{11}\beta_{22} + \beta_{11}\beta_{32} + \beta_{21}\beta_{32} - \beta_{21}\beta_{12} - \beta_{31}\beta_{12} - \beta_{31}\beta_{22})]/Deno; \\ Deno = \alpha_1\alpha_2 + [Hb]_T\{\alpha_1(\beta_{22} + \beta_{32}) + \alpha_2(\beta_{11} + \beta_{21})\} + \\ \qquad [Hb]_T^2(\beta_{11}\beta_{22} + \beta_{11}\beta_{32} + \beta_{21}\beta_{32} - \beta_{21}\beta_{12} - \beta_{31}\beta_{12} - \beta_{31}\beta_{22}); \end{array}\right\}. \tag{A.17}$$

Eqs. (A.15) and (A.16) are spatially averaged over the volume of the RBC, following the method outlined in Appendix C of [18], to obtain the averaged equations given by

$$\frac{d\langle P_1\rangle}{dt} = w_{11}(\langle P_{1,pl}\rangle - \langle P_1\rangle) + w_{12}(\langle P_{2,pl}\rangle - \langle P_2\rangle), \tag{A.18}$$

$$\frac{d\langle P_2\rangle}{dt} = w_{21}(\langle P_{1,pl}\rangle - \langle P_1\rangle) + w_{22}(\langle P_{2,pl}\rangle - \langle P_2\rangle), \tag{A.19}$$

where the coefficients w_{ij} are given as

$$w_{11} = \frac{\eta_1 G_{11}}{b\left(\alpha_1 D_1 + \frac{\eta_1 b}{Sh_i}\right)},$$

$$w_{12} = \frac{\eta_1 G_{12}}{b\left(\alpha_2 D_2 + \frac{\eta_1 b}{Sh_i}\right)},$$

$$w_{21} = \frac{\eta_2 G_{21}}{b\left(\alpha_1 D_1 + \frac{\eta_2 b}{Sh_i}\right)},$$

$$w_{22} = \frac{\eta_2 G_{22}}{b\left(\alpha_2 D_2 + \frac{\eta_2 b}{Sh_i}\right)}. \tag{A.20}$$

Multiplying both sides of Eq. (A.18) by $(\alpha_1 + [Hb]_T\beta_{11})$ and that of Eq. (A.19) by $\{\alpha_2 + [Hb]_T(\beta_{22} + \beta_{32})\}$, we obtain the final form of the spatially averaged forms as in Eqs. (19) and (20), where the diffusing capacities are given by

$$\Theta_{11} = (\alpha_1 + [Hb]_T\beta_{11})w_{11},$$

$$\Theta_{12} = (\alpha_1 + [Hb]_T\beta_{11})w_{12},$$

$$\Theta_{21} = \{\alpha_1 + [Hb]_T(\beta_{22} + \beta_{32})\}w_{21},$$

$$\Theta_{22} = \{\alpha_1 + [Hb]_T(\beta_{22} + \beta_{32})\}w_{22}. \tag{A.21}$$

Plugging in the expressions for w_{ij}s, and D_{ij}s into the above equation system gives the final expressions for the diffusing capacities.

Appendix B. Spatial averaging of CDR equations in the meso scale

Here, we present a brief outline of the Liapunov–Schmidt (LS) reduction of the convection–diffusion equations of the meso-scale. For the sake of mathematical simplicity, we consider equations for only O_2, while those for NO follow a similar treatment. From Eq. (37), we have

$$\frac{1}{\zeta}\frac{d}{d\zeta}\left(\zeta\frac{\partial p_1}{\partial\zeta}\right) = Pe_{T,1}\left\{(1-h)\left(\frac{\partial p_1}{\partial t} + u_{pl}\frac{\partial p_1}{\partial z} - \frac{1}{Pe_1}\frac{\partial^2 p_1}{\partial z^2}\right)\right.$$
$$\left. + h\Theta_{rbc,11}(\langle p_1\rangle - \langle p_{1,rbc}\rangle) + h\Theta_{rbc,12}(\langle p_2\rangle - \langle p_{2,rbc}\rangle)\right\},$$

along with boundary conditions,

$$\left.\frac{\partial p_1}{\partial\zeta}\right|_{\zeta=1} = Pe_{T,1}\frac{\Theta_{1,m}}{2}(1 - p_{1,s}),$$

$$\left.\frac{\partial p_1}{\partial\zeta}\right|_{\zeta=0} = 0,$$

where $p_{1,s} = p_1|_{\zeta=1}$.

We introduce the following definitions of average partial pressure and cup mixing partial pressure, given by

$$\langle p_1\rangle(z,t) = \frac{\int_0^1 p_{1,pl}(\zeta,z,t)2\zeta d\zeta}{\int_0^1 2\zeta d\zeta}, \tag{B.1}$$

$$p_{1,m} = \frac{\int_0^1 p_{1,pl}(\zeta,z,t)u(\zeta)2\zeta d\zeta}{\int_0^1 u(\zeta)2\zeta d\zeta}, \tag{B.2}$$

respectively, where the non-dimensional velocity has the usual parabolic profile for fully developed flow, as mentioned in Eq. (32). We choose to write both pressure and velocity as

$$p_{1,pl} = \langle p_1\rangle(z,t) + p_1'(\zeta,z,t), \tag{B.3}$$

$$u = \langle u\rangle + u'(\zeta), \tag{B.4}$$

where $\langle p_1\rangle$ and $\langle u\rangle$ are spatial averages and p_1' and u' are fluctuations about the average. The fluctuations satisfy the solvability criterion given by

$$\int_0^1 p_1' 2\zeta d\zeta = 0, \tag{B.5}$$

$$\int_0^1 u' 2\zeta d\zeta = 0. \tag{B.6}$$

Plugging these into Eq. (B.2), and using the solvability criterion gives

$$p_{1,m} = \langle p_1\rangle\langle u\rangle + \langle p_1' u'\rangle, \tag{B.7}$$

We note that $\langle u\rangle = 1$, and then proceed to spatially average Eq. (37) over the cross section of the capillary, which gives

$$Pe_{T,1}\Theta_{1,m}(1 - p_{1,s}) = Pe_{T,1}\left\{(1-h)\left(\frac{dp_1}{dt} + \frac{dp_{1,m}}{dz} - \frac{1}{Pe_1}\frac{d^2p_1}{dz^2}\right)\right.$$
$$\left. + h\Theta_{rbc,11}(\langle p_1\rangle - \langle p_{1,rbc}\rangle) + h\Theta_{rbc,12}(\langle p_2\rangle - \langle p_{2,rbc}\rangle)\right\}. \tag{B.8}$$

Using Eqs. (B.3)–(B.6) in Eq. (37), and subtracting the result from Eq. (B.8), we obtain

$$\frac{1}{\zeta}\frac{d}{d\zeta}\left(\zeta\frac{\partial p_1'}{\partial\zeta}\right) = Pe_{T,1}\left[\Theta_{1,m}(1 - p_{1,s})\right.$$
$$-(1-h)\left(\frac{dp_{1,m}}{dz} - \frac{dp_1'}{dt} - u(\zeta)\frac{\partial p_1(\zeta,z,t)}{\partial z} + \frac{1}{Pe_1}\frac{d^2p_1'}{dz^2}\right)$$
$$\left. + h\Theta_{rbc,11}(\langle p_1\rangle - \langle p_{1,rbc}\rangle) + h\Theta_{rbc,12}(\langle p_2\rangle - \langle p_{2,rbc}\rangle)\right]. \tag{B.9}$$

Assuming steady state and considerably high axial Peclet number Pe_1 ($\sim 10^4$, refer to Section 2.2.1.1) allows us to neglect the axial diffusion terms and both Eqs. (B.8) and (B.9) can be further simplified. The simplified form of Eq. (B.8) gives

$$(1-h)\frac{dp_{1,m}}{dz} = \Theta_{1,m}(1 - p_{1,s}) - h\Theta_{rbc,11}(\langle p_1\rangle - \langle p_{1,rbc}\rangle)$$
$$- h\Theta_{rbc,12}(\langle p_2\rangle - \langle p_2, rbc\rangle),$$

which is the same as Eq. (41). This is known as a *Global Evolution Equation*. The simplified form of Eq. (B.9) is given as

$$\frac{1}{\zeta}\frac{d}{d\zeta}\left(\zeta\frac{\partial p_1'}{\partial\zeta}\right) = Pe_{T,1}\left[\Theta_{1,m}(1 - p_{1,s}) - (1-h)\right.$$
$$\left(\frac{dp_{1,m}}{dz} - u(\zeta)\frac{\partial p_1(\zeta,z,t)}{\partial z}\right) + h\Theta_{rbc,11}(\langle p_1\rangle$$
$$\left. -\langle p_{1,rbc}\rangle) + h\Theta_{rbc,12}(\langle p_2\rangle - \langle p_{2,rbc}\rangle)\right]. \tag{B.10}$$

The Liapunov–Schmidt expansion technique then consists of solving Eq. (B.10) for $p_1'(\zeta,z,t)$ by expanding it in powers of the transverse Peclet Number $Pe_{T,1}$ [17], as

$$p_1' = \sum_{j=1}^{\infty}(Pe_{T,1})^j c_j, \tag{B.11}$$

and retaining only the first order terms, which have been shown to be sufficient for qualitative and even quantitative accuracy [17]. Once we calculate p_1', we make use of Eq. (B.7) to obtain the relationship between the modes as

$$p_{1,m} = \langle p_1 \rangle - Pe_{T,1}\left[\frac{\Theta_{1,m}}{24}(1-p_{1,s}) + \frac{(1-h)}{48}\frac{dp_{1,m}}{dz}\right],$$

which is the same as Eq. (43). Similarly, using Eq. (39), we can solve for $p_{1,s}$. The equations thus obtained are called the *Local Equations*. An in-depth development of and discussion on the above may be found in [17].

Appendix C. Derivation of the system of ODEs for solution of coupled micro and meso scale

This section presents the derivation of the "4-ODE system" as outlined in Section 2.4 from the coupled micro and meso scale model (Eqs. (41)–(48)) to produce equations of the form

$$\sum_{i=1}^{n} a_i(x_1, x_2, \ldots x_n)\frac{df(x_1, x_2, \ldots x_n)}{dx_i} = b_i(x_1, x_2, \ldots x_n), \tag{C.1}$$

which can be handled easily by available ODE solver algorithms in softwares such as MATLAB. Algebraic simplification gives the coefficients in Eqs. (60)–(63) as

$$\begin{aligned}
X_1 &= 1 + \frac{3Pe_{T,1}}{8B_1}\Theta_{1,m} + h\Theta_{rbc,11}\left(\frac{Pe_{T,1}}{8B_1} + \frac{\Theta_{1,m}Pe_{T,1}^2}{192B_1}\right),\\
Y_1 &= h\Theta_{rbc,12}\left(\frac{Pe_{T,2}}{8B_2} + \frac{\Theta_{2,m}Pe_{T,2}^2}{192B_2}\right),\\
Z_1 &= \frac{\Theta_{1,m}}{(1-h)}\left(1 - \frac{6p_{1,m} + \Theta_{1,m}Pe_{T,1}}{B_1}\right)\\
&\quad - \left[\begin{array}{l}\Theta_{rbc,11}\left(\frac{3p_{1,m}(8+\Theta_{1,m}Pe_{T,1})+\Theta_{1,m}Pe_{T,1}}{4B_1} - \overline{p_{rbc,1}}\right)\\ +\Theta_{rbc,12}\left(\frac{3p_{2,m}(8+\Theta_{2,m}Pe_{T,2})+k\Theta_{2,m}Pe_{T,2}}{4D_2} - \overline{p_{rbc,2}}\right)\end{array}\right],
\end{aligned} \tag{C.2}$$

$$\begin{aligned}
X_2 &= h\Theta_{rbc,21}\left(\frac{Pe_{T,1}}{8B_1} + \frac{\Theta_{1,m}Pe_{T,1}^2}{192B_1}\right),\\
Y_2 &= 1 + \frac{3Pe_{T,2}}{8B_2}\Theta_{2,m} + h\Theta_{rbc,22}\left(\frac{Pe_{T,2}}{8B_2} + \frac{\Theta_{2,m}Pe_{T,2}^2}{192B_2}\right),\\
Z_2 &= \frac{\Theta_{2,m}}{(1-h)}\left(k - \frac{6p_{2,m} + k\Theta_{2,m}Pe_{T,2}}{B_2}\right)\\
&\quad - \frac{h}{1-h}\left[\begin{array}{l}\Theta_{rbc,21}\left(\frac{3p_{1,m}(8+\Theta_{1,m}Pe_{T,1})+\Theta_{1,m}Pe_{T,1}}{4B_1} - \overline{p_{1,rbc}}\right)\\ +\Theta_{rbc,22}\left(\frac{3p_{2,m}(8+\Theta_{2,m}Pe_{T,2})+k\Theta_{2,m}Pe_{T,2}}{4B_2} - \overline{p_{2,rbc}}\right)\end{array}\right],
\end{aligned} \tag{C.3}$$

$$\begin{aligned}
\overline{X_1} &= -\frac{(1-h)\tau}{u_{rbc}(\alpha_1 + [Hb]_T\beta_{11})}\Theta_{11}\left(\frac{Pe_{T,1}}{8B_1} + \frac{\Theta_{1,m}Pe_{T,1}^2}{192B_1}\right),\\
\overline{Y_1} &= -\frac{(1-h)\tau}{u_{rbc}(\alpha_1 + [Hb]_T\beta_{11})}\Theta_{12}\left(\frac{Pe_{T,2}}{8B_2} + \frac{\Theta_{2,m}Pe_{T,2}^2}{192B_2}\right),\\
\overline{Z_1} &= \frac{(1-h)\tau}{u_{rbc}(\alpha_1 + [Hb]_T\beta_{11})}\left[\begin{array}{l}\Theta_{11}\left(\frac{3p_{1,m}(8+\Theta_{1,m}Pe_{T,1})+\Theta_{1,m}Pe_{T,1}}{4B_1} - \overline{p_{1,rbc}}\right)\\ +\Theta_{12}\left(\frac{3p_{2,m}(8+\Theta_{2,m}Pe_{T,2})+k\Theta_{2,m}Pe_{T,2}}{4B_2} - \overline{p_{2,rbc}}\right)\end{array}\right],
\end{aligned} \tag{C.4}$$

$$\begin{aligned}
\overline{X_2} &= -\frac{(1-h)\tau}{u_{rbc}\{\alpha_1 + [Hb]_T(\beta_{22} + \beta_{32})\}}\Theta_{21}\left(\frac{Pe_{T,1}}{8B_1} + \frac{\Theta_{1,m}Pe_{T,1}^2}{192B_1}\right),\\
\overline{Y_2} &= -\frac{(1-h)\tau}{u_{rbc}\{\alpha_1 + [Hb]_T(\beta_{22} + \beta_{32})\}}\Theta_{22}\left(\frac{Pe_{T,2}}{8B_2} + \frac{\Theta_{2,m}Pe_{T,2}^2}{192B_2}\right),\\
\overline{Z_2} &= \frac{(1-h)\tau}{u_{rbc}\{\alpha_1 + [Hb]_T(\beta_{22} + \beta_{32})\}}\left[\begin{array}{l}\Theta_{21}\left(\frac{3p_{1,m}(8+\Theta_{1,m}Pe_{T,1})+\Theta_{1,m}Pe_{T,1}}{4B_1} - \overline{p_{1,rbc}}\right)\\ +\Theta_{22}\left(\frac{3p_{2,m}(8+\Theta_{2,m}Pe_{T,2})+k\Theta_{2,m}Pe_{T,2}}{4B_2} - \overline{p_{2,rbc}}\right)\end{array}\right].
\end{aligned} \tag{C.5}$$

The above coefficients are plugged into Eqs. (60)–(63), and solved numerically.

References

[1] A. Mansouri, A.A. Lurie, Concise review: methemoglobinemia, Am. J. Hematol. 42 (1993) 7.
[2] M.J. Verive, M. Kumar, Pediatric methemoglobinemia. http://emedicine.medscape.com/article/956528-overview.
[3] R.A. Moodambail, G.R. Pinto, Accidental dapsone induced methemoglobinemia in a young child, Kuwait Med. J. 37 (3) (2005) 207.
[4] R.F. Greer, M. Shanon, Infant methemoglobinemia: the role of dietary nitrate in food and water, Pediatrics 116 (3) (2005) 784.
[5] A.M. Fan, C.C. Willhite, A.S. Book, Evaluation of the nitrate drinking water standard with reference to infant methemoglobinemia and potential reproductive toxicity, Regul. Toxicol. Pharm. 7 (2) (1987) 135.
[6] C.D. Lee, L.K. Ferguson, Methemoglobinemia in emergency medicine. http://emedicine.medscape.com/article/815613-overview.
[7] R.L. Fournier, Basic Transport Phenomena in Biomedical Engineering, Taylor and Francis, 1998.
[8] R.F. Eich, T. Li, D.D. Lemon, D.H. Doherty, S.R. Curry, J.F. Aitken, A.J. Mathews, K.A. Johnson, R.D. Smith, G.N. Phillips Jr., J.S. Olson, Mechanism of NO-induced oxidation of myoglobin and hemoglobin, Biochemistry 35 (22) (1996) 6976–7983.
[9] J.S. Scheele, E. Burner, V.G. Kharitonov, P. Martasek, L.J. Roman, B.S.S. Masters, S.V. Sharma, D. Magde, Kinetics of NO ligation with nitric-oxide synthase by flash photolysis and stopped-flow spectrophotometry, J. Biol. Chem. 274 (19) (1999) 13105.
[10] G.G. Power, S.L. Bragg, B.T. Oshiro, A. Dejam, C.J. Hunter, A.B. Blood, A novel method of measuring reduction of nitrite-induced methemoglobin applied to fetal and adult blood of humans and sheep, J. Appl. Physiol. 103 (4) (2007) 1359.
[11] A. Kinoshita, Y. Nakayama, T. Kitayamat, M. Tomita, Simulation study of methemoglobin reduction in erythrocytes, differential contributions of two pathways to tolerance to oxidative stress, FEBS Journal 274 (6) (2007) 1449.
[12] K. Groebe, G. Thews, Effects of cell spacing and red cell movement upon oxygen release under conditions of maximally working skeletal muscle, Advances in Experimental Medicine and Biology 248 (1989) 175.
[13] N.S. Huang, J.D. Hellums, A theoretical model for gas transport and acid/base regulation by blood flowing in microvessels, Microvasc. Res. 48 (1994) 364.
[14] J.D. Hellums, P.K. Nair, N.S. Huang, N. Oshima, Simulation of intraluminal gas transport processes in the microcirculation, Ann. Biomed. Eng. 24 (1996) 1.
[15] R.W. Flummerfelt, E.D. Crandall, An analysis of external respiration in man, Math. Biosci. 3 (1968) 205.
[16] R.E. Forster, E.D. Crandall, Pulmonary gas exchange, Annu. Rev. Physiol. 38 (1976) 69.
[17] S. Chakraborty, V. Balakotaiah, Averaging theory and low dimensional models for chemical reactors and reacting flows, Chem. Eng. Sci. 58 (2003) 4769.
[18] S. Chakraborty, V. Balakotaiah, A. Bidani, Diffusing capacity reexamined: relative roles of diffusion and chemical reaction in red cell uptake of O_2, CO, CO_2 and NO, J. Appl. Physiol. 97 (2004) 2284.
[19] S. Chakraborty, V. Balakotaiah, A. Bidani, Multiscale model for pulmonary oxygen uptake and its application to quantify hypoxemia in hepatopulmonary syndrome, J. Theor. Biol. 244 (2007) 190.
[20] T. Kagawa, M. Mochizuki, Numerical solution of partial differential equation describing oxygenation rate of red blood cell, Jpn. J. Physiol. 32 (1982) 197.
[21] P.K. Nair, J.D. Hellums, J.S. Olson, Prediction of oxygen transport rates in blood flowing in large capillaries, Microvasc. Res. 38 (1989) 269.
[22] J.B. West, Ventilation/Blood Flow and Gas Exchange, Blackwell Scientific Publications, Oxford, Boston, 1977.
[23] J.W. Severinghaus, Simple, accurate equations for human blood oxygen dissociation computations, J. Appl. Physiol. 46 (1979) 599.
[24] E.A. Lissi, T. Caceres, Oxygen diffusion-concentration in erythrocyte plasma membranes studied by the fluorescence quenching of anionic and cationic pyrene derivatives, J. Bioenerg. Biomembr. 21 (3) (1989) 375.
[25] A. Denicola, J.M. Souza, R. Radi, E. Lissi, Nitric oxide diffusion in membranes determined by fluorescence in quenching, Arch. Biochem. Biophys. 328 (1) (1996) 208.
[26] American Thoracic Society (ATS) documents, Statement on home care for patients with respiratory disorders, Am. J. Respir. Crit. Care Med. 171 (2005) 1443.